5-2-2: Examining Software-based Tools

At the end of this episode, I will be able to:

1. Identify and explain the role of software-based network troubleshooting tools, given a scenario.

Learner Objective: *Identify and explain the role of software-based network troubleshooting tools, given a scenario*

Description: In this episode, the learner will examine various software-based tools used in network troubleshooting. We will explore protocol analyzers, packet capture utilities, Wi-Fi Analyzers, and more.

- If we need to look at the network traffic, what software can we use?
    - Protocol analyzers:
        - Protocol analyzers are tools designed to decode and analyze network traffic to understand protocols' behavior, diagnose problems, and ensure network security.
    - Packet capture software:
        - Packet capture software focuses on capturing and recording network traffic, allowing for detailed examination and analysis at a later time.
- How about in wireless networks, what might we use?
    - Wi-Fi analyzers
        - A type of tool that can use wireless networks to diagnose performance and configuration issues
            - Assess signal strength
            - Identify and remediate congestion and interference issues
            - Verify and optimize channel use
- Are websites that provide speed tests beneficial?
    - Speed test websites:
        - These websites measure Internet connection speed
        - They test the download and upload rates
        - Additional options, such as geolocation or server choice.
        - They can provide insights into network performance and quality.
- How about some "honorable mentions" for software-based tools?
    - *nmap*
        - Network Mapper, perform network scans
        - Command line interface or CLI
    - *tcpdump*
        - Dump/capture packets off the network
        - Command line interface or CLI

- Additional Resources:
    - Wireshark display filters

```
ip.src == 192.168.0.10
tcp[13]&2!=0 (ID TCP three-way handshake)
tcp.port eq 443 or arp (display port 443, or ARP)
smb || nbns || dcerpc || nbss || dns (filter based on common Windows domain cont
```

    - Wi-Fi Analyzers
        - Signal strength of approximately -30 dBm to -50 dbm is excellent signal strength, with 60 dBm being fair and above considered poor

        - ♠ Consider AP and antenna position, and coverage area
    - ♠ Channel overlap - look for APs on the same channel, if necessary manually change channel
    - ♠ AP density - the more APs broadcasting in a location, the greater the potential for interference
    - ♠ Configuration - verify the AP and/or client are running the same communication, encryption and authentication protocols.
- o Speed Test websites
    - ♠ https://www.speakeasy.net/speedtest/
    - ♠ https://www.speedtest.net/
    - ♠ https://fiber.google.com/speedtest/